

Differential Calculus — Chain Rule

Chapter 2: Differentiation of Composite Functions | Page 33

PROBLEM STATEMENT

Power Rule & Linear Composite

Find the derivative with respect to x of the function:

$$y = (5x - 3)^4$$

STEP 1 প্রদত্ত বহুপদী ঘাত ফাংশনটি উল্লেখ করি (Given Composite Function)

$$y = (5x - 3)^4$$

STEP 2 উভয়পক্ষে x -এর সাপেক্ষে অন্তরীকরণ করি (Differentiating both sides w.r.t. x)

$$\frac{dy}{dx} = \frac{d}{dx} [(5x - 3)^4]$$

STEP 3 শক্তি বিধি ও চেইন রুল প্রয়োগ করি ($\frac{d}{dx}[u^n] = nu^{n-1} \frac{du}{dx}$)

$$\frac{dy}{dx} = 4(5x - 3)^{4-1} \cdot \frac{d}{dx}(5x - 3) = 4(5x - 3)^3 \cdot \frac{d}{dx}(5x - 3)$$

STEP 4 অভ্যন্তরীণ রৈখিক অংশের অন্তরক নির্ণয় করি (Differentiating Inner Linear Part)

$$\frac{d}{dx}(5x - 3) = 5 \frac{d}{dx}(x) - \frac{d}{dx}(3) = 5(1) - 0 = 5$$

STEP 5 মান বসিয়ে ধ্রুবকগুলোর গুণফল হিসাব করে চূড়ান্ত অন্তরক পাই (Final Evaluation)

$$\frac{dy}{dx} = 4(5x - 3)^3 \times 5 = 20(5x - 3)^3$$

FINAL ANSWER $\frac{dy}{dx} = 20(5x - 3)^3$

Rigorous Insights & Mark Scheme

AMIE Section-A Standard Evaluation Protocol | Page 33, HW 01

BOX 1: FORMULATION & GENERALIZED POWER RULE

- **Generalized Power Rule:** $\frac{d}{dx}[f(x)]^n = n[f(x)]^{n-1} \cdot f'(x)$.
- **Linear Chain Rule Property:** For any linear composition $(ax + b)^n$, the derivative is simply:

$$\frac{d}{dx}(ax + b)^n = na(ax + b)^{n-1}$$

- Here, $a = 5$, $b = -3$, and $n = 4 \implies \frac{dy}{dx} = 4 \times 5 \times (5x - 3)^3 = 20(5x - 3)^3$.

BOX 2: COMMON STUDENT TRAPS & PITFALLS

- **Missing Inner Derivative:** Writing $\frac{dy}{dx} = 4(5x - 3)^3$ by ignoring $\frac{d}{dx}(5x - 3) = 5$. This is the most common student omission in chain rule questions.
- **Unnecessary Expansion:** Attempting to expand $(5x - 3)^4$ into a full polynomial via Binomial Theorem. This wastes critical exam time and increases algebraic error risk.

BOX 3: REAL DOMAIN & DIFFERENTIABILITY ANALYSIS

- **Domain of Definition:** The polynomial function $y = (5x - 3)^4$ is smooth and defined for all real numbers $x \in \mathbb{R}$.
- **Class of Differentiability:** $y \in C^\infty(\mathbb{R})$ (infinitely differentiable everywhere).
- **Stationary Point / Critical Point:** Setting $\frac{dy}{dx} = 0 \implies 5x - 3 = 0 \implies x = \frac{3}{5}$. At $x = 0.6$, the curve attains a global minimum with $y = 0$.

BOX 4: LEIBNIZ SUBSTITUTION METHOD (ALTERNATIVE RIGOR)

- Let the intermediate variable be $u = 5x - 3$, so that $y = u^4$.
- Outer derivative: $\frac{dy}{du} = \frac{d}{du}(u^4) = 4u^3$.
- Inner derivative: $\frac{du}{dx} = \frac{d}{dx}(5x - 3) = 5$.
- Chain Rule multiplication:

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = 4u^3 \cdot 5 = 20u^3 = 20(5x - 3)^3$$

BOX 5: AMIE EXAMINATION MARKS DISTRIBUTION (TOTAL: 3 / 4 MARKS)

- **1.0 Mark:** Correct identification and citation of the generalized power rule formula.
- **1.0 Mark:** Accurate calculation of the inner linear derivative $\frac{d}{dx}(5x - 3) = 5$.
- **1.0 Mark:** Correct algebraic simplification to obtain $20(5x - 3)^3$ with clear framing.